



MLE4228 AY25/26 Semester 1 Lab 2 Report

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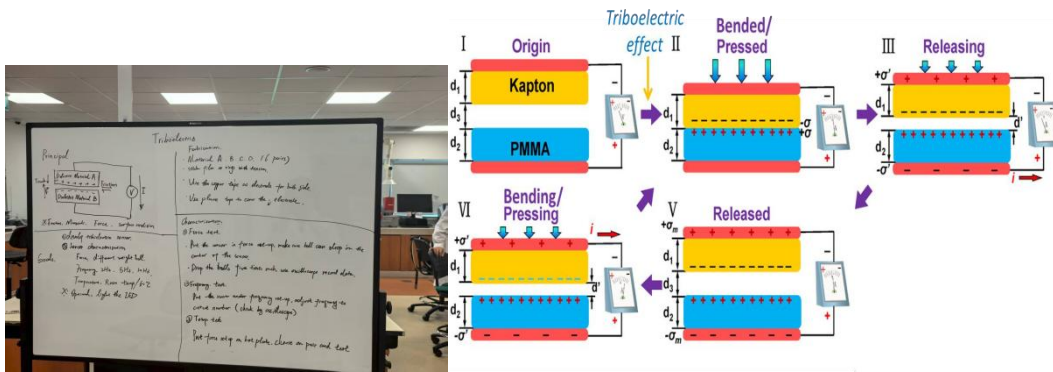
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1.INTRODUCTION

Sensors play a critical role in modern technology, compared to conventional sensors, triboelectric sensors are lightweight, low-cost. This type of sensor uses friction between materials to generate and detect electrical signals.

The triboelectric sensor operates based on the triboelectric effect and electrostatic induction. When two dissimilar materials come into contact and separate, surface charge transfer occurs, generating a potential difference that can be measured as an electrical output. This kind of sensor is energy-independent and is widely in wearable electronics for receiving human instructions in human-machine interfaces for motion detection, which is important in soft robots.

In this experiment, six sensors were fabricated and tested. The aim is to find the difference between the six sensors, test their performance under mechanical stimulation. The experiment also shows how variations in applied force and frequency influence the electrical output, and to demonstrate the potential of triboelectric devices as practical, self-powered sensors.



2.OBJECTIVE

✓ Test Material Sensitivity

The material sensitivity of six sensor samples, fabricated from 4 different material combinations, was evaluated by applying forces of varying magnitudes.

✓ Force Response and Linearity Testing

The most responsive sample was selected for force response and linearity testing, where the output signal was recorded and analyzed for its relationship with applied force. The frequency response and repeatability of the sensor were then examined by subjecting it to mechanical inputs at different frequencies.

✓ Frequency Response and Repeatability Testing

The frequency response and repeatability of the sensor were then examined by subjecting it to mechanical inputs at different frequencies.

✓ Sensitivity to Interference

The sensor's sensitivity to external interference was investigated by testing its performance under heat conditions.

3.MATERIAL & METHODS



● Material

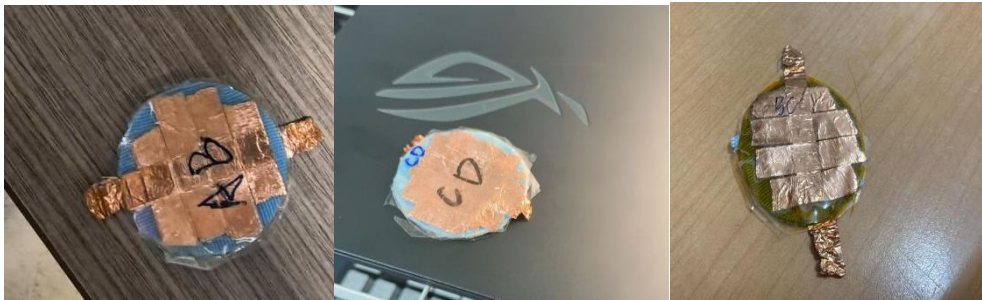
As shown in the figure, four different materials (labeled from A to D) with distinct physical properties were provided for the fabrication of triboelectric sensors. By combining these materials in pairs, six different sensor samples (Sample AB to Sample CD) were prepared.

The materials varied in terms of flexibility, transparency, and surface texture, which allowed for comparative testing of their triboelectric performance. Material A, B are transparent and C, D are respectively orange and white. Except for A which is in a film-like form, the rest are all hard sheet-like materials.

A standard 3D-printed setup, a frequency response setup, some plastic blue 3D printed separator, double sides adhesive tape (both insulating and conductive), an oscilloscope with USB, a set of metal balls of varying weights, a multimeter, a digital scale, and a hotplate were also used in the experiments.

- Fabrication

- a. Four different materials were arranged and combined to fabricate six types of sensors for testing. First, material were cut into small round and stick to the blue 3D printed separator. Different materials were pasted on both sides of the separator, the materials are fully stretched before stuck to the separator.
- b. Conductive sticker were covered on the surface of material of both sides, which resulted in the formation of a capacitor. Longer strips of conductive tape on both sides were left to facilitate connection with the oscilloscope.
- c. After covering the conductive surface by scotch tape to protect the sensor, each sensor were test by the multimeter. In the experiment, it is necessary to ensure that there is insulation between the two sides of the sensor, while one side is conductive.



- Experimental setup and method

- ✓ Task 1- Material Sensitivity:

6 samples were connected to the oscilloscope. We dropped metal ball of four different size (ball's size: 80.00g, 44.09g, 16.93g, 9.09g) from a height of 15 cm onto 6 samples using the 3D printed setup. After recording the output signal to the USB we repeat the experiment for each one for 5 times to minimize the possible error.

- ✓ Task 2- Force Response and Linearity Testing:

The most responsive sample observed in previous step was selected. We used balls of different weights and drop each one onto the selected sample to test its pressure/force-based response. The output signal was recorded to the USB, which enabled us to analyse the linearity of the sensor.

- ✓ Task 3- Frequency Response and Repeatability Testing:

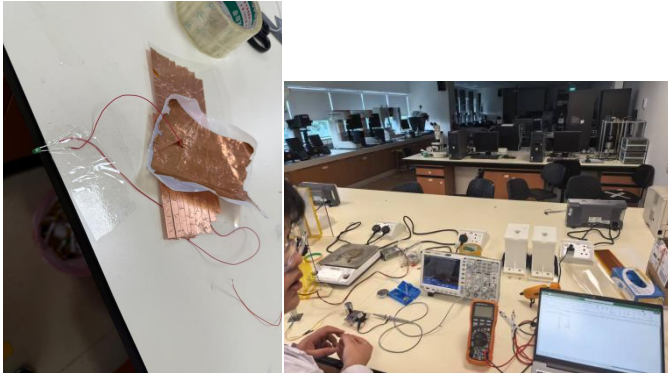
The most responsive sample from material sensitivity testing was selected. We used the frequency response setup to measure its response under a fixed force and recorded data for frequency response at 2 Hz, 5 Hz, and 10 Hz. This data will also be used to analyze repeatability.

- ✓ Task 4- Sensitivity to Interference:

To test the performance of the triboelectric sensor under different external parameters (heat), we place the sensor on a hotplate set to 50°C and record any changes in signal output at the same frequency, saving the data to the USB.

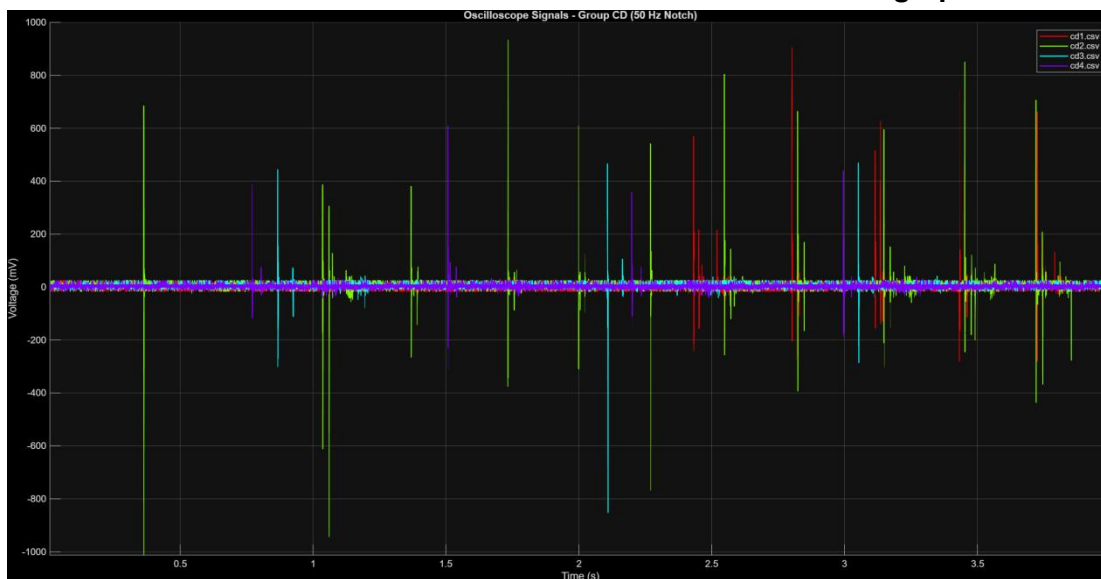
✓ Task 5- Self-powered robotics sensor demo:

The LED was powered by the force setup using highest performance frequency point observed in previous steps. We have done some adjustment to sensor to increase power (increase sensor area and increase friction)



4.RESULTS

➤ **Detect how the sensors work and the characteristics of the graph in material sensitivity test:**



The triboelectric sensor demonstrated several key characteristics:

1. **Force Responsiveness** – The sensor output responds directly to external mechanical stimuli (e.g., pressing, tapping, or sliding). Any applied force generates a measurable electrical signal. It was able to detect changes in external force.
2. **Peak signal shape** – Each contact-and-release cycle produces a bipolar output:
 - A positive peak when the surfaces come into contact or pressure increases.
 - A negative peak of similar magnitude when the surfaces separate or pressure decreases.
 - Ideally, for single-touch detection, the target is to isolate or filter one dominant peak.

The graph was made by MATLAB.

➤ **Use the most responsive sample observed in previous step to do the force response and linearity testing:**

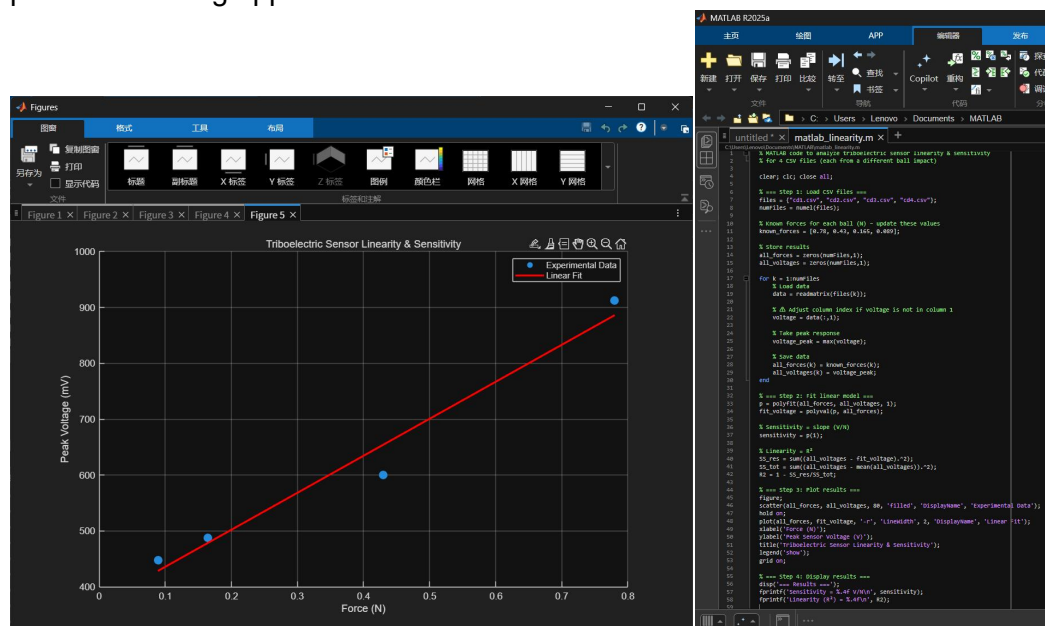
(During the experiment, we found that CD is the most responsive sensor. As the result, CD was used in the following experiment. The comparison of the six sensors will be explain soon in Discussion part.)

=== Results ===

Sensitivity = 662.2332 mV/N

Linearity (R^2) = 0.96

The triboelectric sensor exhibited a sensitivity of 662.2 mV/N, indicating a strong voltage response to applied force. The linearity of the force–output relationship was high, with an R^2 value of 0.9691, showing that the sensor's response was highly consistent and predictable across the tested range. These results suggest that the sensor demonstrates both good sensitivity and excellent linearity, making it suitable for practical sensing applications.



The experimental data from four CSV files were analyzed using MATLAB.

Since each CSV file only contains the voltage response of the sensor under the impact of a different ball, the maximum voltage peak from each dataset was extracted to represent the sensor's response to that particular impact.

Known force values for the four balls were assigned based on their physical properties. (ball's size: 80.00g, 44.09g, 16.93g, 9.09g)The corresponding voltage peaks and forces were then paired to form a force–voltage dataset. MATLAB performed a linear regression (polyfit) to obtain the slope of the line, which represents the sensitivity of the sensor (in V/N).

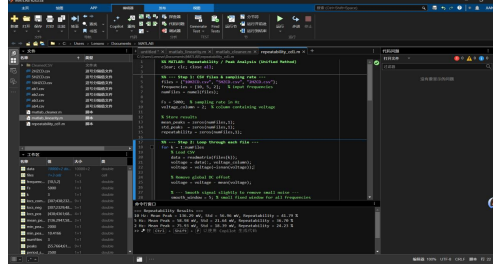
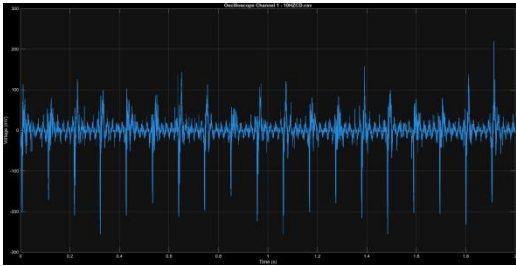
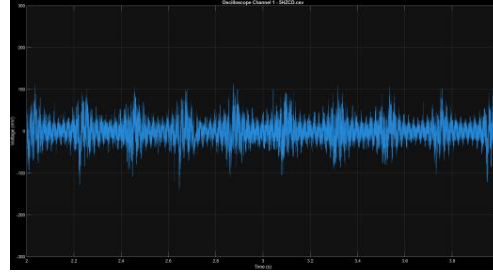
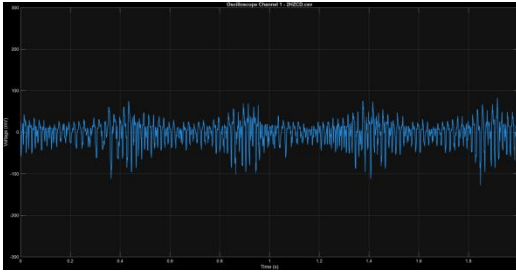
To evaluate linearity, the coefficient of determination (R^2) was calculated by comparing the measured data points against the fitted line. The results were also visualized in a scatter plot (experimental data) with the best-fit line overlaid to show the degree of linearity.

➤ Find out frequency response in repeatability testing:

Repeatability (%) is a measure of how consistent a sensor's response is when it is subjected to repeated, identical impacts under the same conditions.

Mathematically:

Repeatability= (Standard Deviation of Peak Values/ Mean peak value)*100%



=== Repeatability Results ===

10 Hz: Mean Peak = 106.57 mV, Std = 20.58 mV, Repeatability = 28.84 %

5 Hz: Mean Peak = 72.61 mV, Std = 15.15 mV, Repeatability = 20.87 %

2 Hz: Mean Peak = 32.97 mV, Std = 7.95 mV, Repeatability = 18.28 %

The repeatability of the triboelectric sensor was analyzed for three input frequencies: 2 Hz, 5 Hz, and 10 Hz.

- For the 10 Hz input, the sensor exhibited a mean peak of 106.57 mV with standard deviation of 20.58 mV, resulting in a repeatability of 28.84 %, indicating moderate variability in response due to the rapid impacts.
- At 5 Hz, the mean peak was 72.61 mV with a standard deviation of 15.15 mV, yielding a repeatability of 20.87 %, showing a more consistent response compared to 10 Hz.
- The 2 Hz input produced the lowest mean peak of 32.97 mV with a standard deviation of 7.95 mV, giving a repeatability of 18.28 %, demonstrating the highest stability for slow impacts.

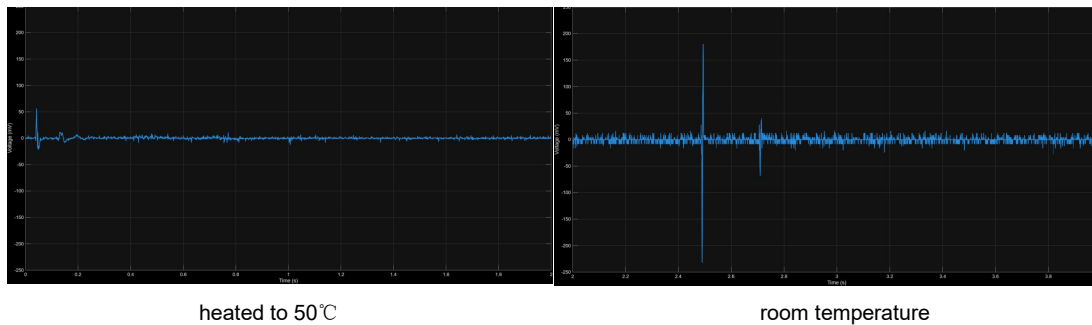
Overall, the results indicate that faster impacts generate higher peak voltages but slightly less consistent responses, whereas slower impacts produce lower voltages with greater stability. This analysis confirms that the sensor's output is reliable and predictable, and repeatability provides a quantitative metric to assess its precision across different impact frequencies.

The experimental data from three CSV files, each corresponding to a different input frequency (2 Hz, 5 Hz, and 10 Hz), were analyzed using MATLAB. Each CSV file contained the time-domain voltage response of the triboelectric sensor under repeated impacts at a fixed frequency. To reduce high-frequency noise and remove baseline drift, the raw signals were pre-processed using a smoothing function and detrending. The primary voltage peaks corresponding to each impact were then identified using MATLAB's findpeaks function, with the minimum peak distance set according to the expected impact period to ensure that only one peak was counted per cycle.

For each frequency dataset, the absolute voltage peak values from all impacts were collected. MATLAB then computed the mean peak amplitude, the standard deviation of the peaks, and the repeatability percentage, defined as the ratio of standard deviation to mean amplitude multiplied by 100.

➤ **Find out sensitivity to interference**

During the heat test, the sample was heated to 50 degree celsius. Compared to room-temperature measurements, the output peak voltage decreased significantly. This reduction confirms that high temperature interferes with the triboelectric charge generation process, lowering the effective sensitivity of the sensor under identical mechanical excitation.



➤ **Result of the self-powering test (LED)**

The LED was successfully powered by the triboelectric sensor. Although the emitted light was relatively dim and fluctuated between illumination and decay, this result confirms the sensor's self-powering capability. However, the generated current was relatively small and unstable, limiting the brightness and consistency of the LED.

5.DISCUSSION

➤ **Fabrication process, sample preparation, and any challenges encountered.**

During the fabrication process, two major challenges were encountered.

First, controlling the thickness of the adhesive tape was difficult. If the tape overlapped, it made the surface uneven and changed the spacing between the two materials, thereby affecting the sensor's performance. To address this issue, the tape was carefully trimmed to match the exact dimensions of the separator before being applied.

Second, it was essential to ensure the presence of a uniform gap between the two materials. To achieve this, both materials were gently stretched and tested using a multimeter. Successful insulation confirmed that the gap had been properly formed.

➤ **Compare the performance of different sensor materials.**

When we compared the six triboelectric sensors, we needed objective metrics to evaluate their performance. We used three key criteria:

- **Sensitivity(G)**

Defined as the average output voltage response (Mean_VPP) to external stimuli. A higher sensitivity means the sensor can detect smaller forces or impacts more effectively.

Ranking rule: Higher Mean_VPP → better → lower rank number.

- **Stability(H)**

Defined as the average background noise level (Mean_Noise) when no external force is applied. A stable sensor should have minimal fluctuations in the baseline signal.

Ranking rule: Lower Mean_Noise → better → lower rank number.

- **Linearity(I)**

Defined by the regression fit (R^2 value) between ball size (impact strength) and sensor output. (ball's size: 80.00g, 44.09g, 16.93g, 9.09g)

A linear response indicates that the sensor can reliably scale its output with the applied force.

Ranking rule: Higher $R^2 \rightarrow$ better $\rightarrow$ lower rank number.

- Overall Rank(J)

We summed the three ranks (sensitivity, stability, linearity) to give each sensor an Overall Rank.

The sensor with the lowest total rank(CD) is considered the **best overall performer**.

	A	B	C	D	E	F	G	H	I	J	K
1	Sensor	Mean_Vpp	Mean_Noise	Baseline	Slope	R2	Sensitivity	Stability	Repeatability	Overall Rank	
2	AB	0.361531	0.00414	-0.00016	-0.1864	0.91737	5	1	4	3	
3	AC	0.376722	0.007868	-0.00046	-0.10948	0.947376	4	3	3	3	
4	AD	0.619901	0.013761	0.000121	-0.20102	0.872007	3	5	6	6	
5	BC	0.22676	0.004984	-0.00064	-0.03253	0.976749	6	2	2	3	
6	BD	0.939437	0.044817	-0.00458	-0.37771	0.911563	2	6	5	5	
7	CD	0.994243	0.011555	0.003095	-0.1436	0.97942	1	4	1	1	
8											
9											

- Working Principle of the Python Analysis

We analyzed your 24 CSV files (6 sensors $\times$ 4 ball sizes) in a systematic pipeline:

1.Data Import

Each CSV file contains a time series signal:

Column 1 = time

Column 2 = voltage output of the triboelectric sensors

Files were grouped: e.g. ab1.csv ... ab4.csv for sensor AB, cd1.csv ... cd4.csv for sensor CD.

2. Noise Filtering

We removed the most common interference frequencies:

50 Hz $\rightarrow$ typical power line noise

10 Hz $\rightarrow$ environmental/mechanical vibration

Used bandstop filters (notch filters) in Python (scipy.signal.iirnotch + filtfilt) to clean the signals.

3. Metric Extraction (per file)

For each signal (sensor $\times$ ball size), we calculated:

- Vpp (Peak-to-Peak Voltage)
- Vpp=max(signal)-min(signal)
 $\rightarrow$ Measures sensitivity (how strong the output is when hit).
- Baseline (Mean Value)
Average signal when no ball is hitting.
- Noise (Standard Deviation of Baseline)
Measures stability (how "quiet" the sensor is at rest).

4.Sensor-Level Summary (per sensor)

Across all 4 ball sizes, we computed:

Mean_Vpp = average Vpp $\rightarrow$ sensitivity indicator

Mean_Noise = average noise $\rightarrow$ stability indicator

Slope & R^2 from regression (ball size $\rightarrow$ Vpp) $\rightarrow$

If Vpp grows consistently with ball size, slope is large and $R^2 \approx 1$.

Measures linearity.

➤ Analyze and discuss the following aspects of the triboelectric sensors: sensitivity, repeatability, linearity, frequency response, and sensitivity to interference.

1. Sensitivity

The sensitivity of the sensor was calculated as the slope of the output voltage versus applied force curve. The obtained sensitivity was 662.2 mV/N, indicating that for every 1 N increase in force, the output voltage increases by approximately 662.2 mV.

This value suggests that the sensor (CD) is relatively high compared to other samples, which determine the application of the sensor.

- Why some sensors were more or less sensitive.

Main factor: Material choice

Different materials have different tendencies to gain or lose electrons when in contact. If one layer is strongly positive and the other strongly negative (from the triboelectric series), larger charge separation → higher sensitivity. Materials with higher electron affinity differences lead to bigger signal outputs. Just like C and D in the experiment.

Other Factors: surface roughness, contact area, thickness of layers, the higher sensitivity can be attributed to smoother surface contact, higher surface charge density and so on.

- Comment on practical implications: e.g., higher sensitivity → better detection of small forces, but may saturate under high loads.

Higher sensitivity in a triboelectric sensor allows it to detect smaller forces or subtle mechanical stimuli, making it ideal for applications such as wearable health monitors, tactile sensing, or pressure mapping. However, extremely high sensitivity may also have drawbacks: under large forces, the sensor can saturate, meaning the output voltage no longer increases proportionally with applied force, reducing accuracy at higher loads.

2. Repeatability

To evaluate repeatability, the sensor was tested under identical loading conditions for multiple cycles (2Hz, 5Hz, 10Hz). This demonstrates that the sensor CD shows good repeatability in this experiment, meaning it can consistently reproduce results under the same input. Any variation observed in may result from contact wear, surface contamination, or uneven pressure distribution.

- Discuss any possible causes of fluctuation in the experiment
alignment issues, uneven surface, wear of materials, environmental noise.

- Discuss whether your repeatability is acceptable for intended applications.

The repeatability of the sensor CD indicates a moderate level of consistency. For applications such as low-precision force detection or general motion sensing, this level of repeatability may be acceptable, as the sensor can reliably detect changes in force trends. However, for high-precision applications, such as medical monitoring or tactile feedback devices, the variability may be too high, potentially introducing errors.

3. Linearity

The linearity of the sensor response was assessed by fitting the output–force curve to a linear regression. The coefficient of determination was $R^2 = 0.96$ (CD). A value close to 1 indicates a strong linearity (As Cd did), meaning the sensor output is directly proportional to applied force. Minor deviations from linearity may be due to saturation at higher forces, material deformation, or irregular contact and noise interference.

- Interpret R^2 values: good linearity → easier calibration, predictable output.
- Discuss why strong linearity was observed in this experiment.

Factors contributing to this linear behavior include uniform surface contact, stable material properties, and controlled fabrication, which minimized irregularities that could cause deviations. Strong linearity is advantageous for calibration, as it ensures that the sensor output can be reliably converted to force or pressure values.

4. Frequency Response

The sensor was tested at different input frequencies (2 Hz, 5 Hz, 10 Hz). The average peak varied when the voltage changed.

The results show that the output increases with frequency. . The sensor is therefore suitable for high frequency applications.

- Discuss how and why output changes with frequency.

In this experiment, the sensor's mean output voltage increased with higher input frequencies. This behavior can be explained by faster contact-separation leading to incomplete charge transfer, or enhanced charge accumulation at higher frequencies. Also faster contact-separation cycles can generate more frequent charge transfers per unit time, leading to a higher average voltage.

5. Sensitivity to Interference

The stability of the sensor was tested under potential interference condition (heated). The output showed significant fluctuations, suggesting that the sensor CD is highly sensitive to environmental disturbances. To improve performance, shielding, encapsulation, or signal filtering could be applied.

- Discuss why heat will affect the sensor's performance

Heat can significantly affect triboelectric sensor performance because temperature influences the material properties and charge transfer processes. At higher temperatures, the dielectric constant and surface conductivity of the materials may change, which can lead to **faster charge dissipation** and lower net voltage output.

- Suggest improvements: shielding, packaging, filtering, or material modifications.

➤ **Evaluate the self-powering performance of the triboelectric sensor, particularly in the context of powering a LED (if applicable).**

Since a LED typically requires a forward voltage of approximately 2 V, the sensor's output was compared against this threshold. The self-powering performance of the triboelectric sensor in this experiment is moderate. Although the sensor produces enough voltage, the current output may be too low for stable operation.

- Discuss the reason why the self-powering performance is moderate.

although it can generate voltage through the triboelectric effect, the current output is limited

- Discuss how to improve the performance.

Tough material performed better than soft material.

➤ **Unexpected results or discrepancies observed during the experiments.**

During the experiment, AB actually showed good performance in the material sensitivity test (sometimes even better than CD), which caused a lot of difficulties when the best sensor should be picked up. However, when importing AB's csv data into MATLAB code, AB's performance is not as good as CD's.

- Discuss the reason of the phenomenon.

This discrepancy may be due to measurement noise. We noticed that the peak value of AB is often not a single peak, but rather five peaks within a small grid (each grid representing 0.1 seconds). This seems to indicate that the 50Hz noise significantly affects the reading of AB. We suspect that this is due to the unstable connection of the circuit during each impact. Additionally, transient spikes in the raw data might have temporarily inflated AB's apparent performance

- Discuss how to solve the problem.

Add 50HZ filter, smooths out these fluctuations, revealing the true average performance.

➤ **Reflect on the challenges encountered and propose potential solutions.**

Misidentified the noise as the target peak.

Solutions: A 50 Hz notch filter was applied in the MATLAB code to remove the most common source of electrical noise, ensuring that only the true sensor peaks were detected.

- Unreasonable data appeared in the outcome of MATLAB.

Solution: delete improper smoothing code, peak detection threshold too high or inappropriate, ignoring negative peaks, low-frequency segmentation errors

6. CONCLUSIONS

➤ **Summarize the key findings.**

- a. From the material sensitivity test and overall data analysis, sample CD demonstrated the best performance among all tested sensors, particularly in terms of sensitivity, stability, and linearity.
- b. The linearity and sensitivity results confirmed this: CD achieved an excellent linearity ($R^2 = 0.96$) and a high sensitivity (662.2 mV/N), while the other samples showed only moderate performance.
- c. Frequency and repeatability tests revealed that CD's repeatability was moderate, indicating some variability in output across repeated trials.
- d. Environmental testing showed that the triboelectric sensors were highly sensitive to external conditions, as heating significantly altered their performance compared to room temperature operation.
- e. Self-powering test showed that triboelectric sensor can light up the LED but the current was not stable.

➤ **Discuss whether the objectives were addressed.**

Overall, the experiment objectives were largely achieved. Due to the limited time, we were unable to complete the experiment of Sliding Mode Triboelectric Sensor. Some discussions and most of the data processing were conducted after the lab session.

➤ **Evaluate the overall success of the experiment.**

Overall, the experiment was successful. We not only achieved most of the experimental objectives but also developed valuable skills in analyzing and interpreting the obtained data, such as MATLAB and python.

➤ **Suggest further research directions or improvements**

1. Future work could improve sensor repeatability and reduce environmental sensitivity. Exploring protective packaging and testing with different materials may also enhance performance.
2. Application testing → integrating the sensor into real devices (wearables, touchpads, robotics).
3. Fabrication improvements → better control of surface roughness or thickness.

7. REFERENCE

- 1, Wang, Z.L. (2017). *Triboelectric Nanogenerators as New Energy Technology and Self-Powered Sensors – Principles, Concepts, and Applications*. Advanced Energy Materials, 7(24), 1701180.
2. Zhu, G., et al. (2014). *Triboelectric Nanogenerators as New Energy Technology for Self-Powered Systems and Sensors*. Nano Energy, 14, 161–174.